

The exact seven-product closure degree is forty-seven

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17 September 2026

Abstract

Starting from $1, x$ and allowing arbitrary complex linear combinations for free, we determine the largest degree whose entire coefficient space lies in the Zariski closure of outputs computable with seven products. The answer is 47. A 49-parameter polynomial family has an exactly certified contact of order 47 with a nonsingular 48-variable coefficient differential. An inverse-function and scaling argument gives coverage in the full degree-128 ambient space. A self-contained dimension bound, together with irreducibility, proves that degree 48 cannot be covered. The computational certificate uses rational arithmetic and has been independently reconstructed.

Assistance and review. This work was developed with substantial ChatGPT and Codex assistance. Separate AI agents reviewed the mathematical argument and independently implemented the certificate checks. Their reports and executable evidence accompany this manuscript. These are informal AI-agent reviews, not external human peer review or proof-kernel formalization.

1 Statement and computational model

Write $V_d = \mathbb{C}[x]_{\leq d}$. Begin with the scalar polynomials $1, x \in \mathbb{C}[x]$. Arbitrary complex linear combinations of available polynomials are free; forming a product of two available polynomials costs one multiplication. Let $\mathcal{P}_7 \subseteq V_{128}$ be the possible outputs using at most seven products, and define

$$X_7 = \overline{\mathcal{P}_7}^Z \subseteq V_{128}, \quad d_7 = \max\{d \in \{0, \dots, 128\} : V_d \subseteq X_7\}. \quad (1)$$

The closure is taken in all 129 coefficients. In particular, an approximating output may have higher degree than its limit. Discarding high-degree coefficients is not a permitted computational operation.

Theorem 1. *In the model (1), one has $d_7 = 47$.*

MF-14 and Conjecture 13 of Jarlebring–Lorentzon [1] asked whether this maximum equals 42. Colbrook [2] proved the degree-42 coverage statement with an exact coefficient-rank certificate. The author’s earlier degree-44 construction [3] refuted the equality to 42 and left $44 \leq d_7 \leq 47$. Theorem 1 determines the exact maximum. Its proof below is independent of those lower-bound constructions and requires no auxiliary theorem for a smaller multiplication budget.

The assertion concerns complex coefficient-space closure. It does not assert exact representation of every degree-47 polynomial, real coverage, or numerical stability of coefficient recovery. Substituting a matrix for x gives the corresponding matrix-polynomial evaluation model, in which matrix products dominate the cost.

2 A polynomial family in the seven-product closure

Let $\alpha, b, c, \eta \in \mathbb{C}$ and set

$$q_1 = x^2, \quad q_2 = Q = x^4 + \alpha x^3, \quad q_3 = R = Q^2 + bx^2Q + cxQ + \eta x^3. \quad (2)$$

For $k = 4, 5, 6, 7$, introduce the ordered list

$$L_k = (r_0, \dots, r_{k-2}) = (x, q_1, \dots, q_{k-2})$$

and define

$$q_k = \left(q_{k-1} + \sum_{j=0}^{k-2} u_{k,j} r_j \right) \left(q_{k-1} + \sum_{j=0}^{k-2} v_{k,j} r_j \right). \quad (3)$$

Finally, put

$$\Phi(\theta) = e_0 + e_1 x + \sum_{k=1}^7 e_{k+1} q_k. \quad (4)$$

There are $4 + (6 + 8 + 10 + 12) + 9 = 49$ parameters. Their order is

$$\theta = (\alpha, b, c, \eta, \mathbf{u}_4, \mathbf{v}_4, \mathbf{u}_5, \mathbf{v}_5, \mathbf{u}_6, \mathbf{v}_6, \mathbf{u}_7, \mathbf{v}_7, e_0, \dots, e_8), \quad (5)$$

with each vector ordered by increasing j . Table 1 specifies the zero-based indices used by the certificate. The map $\Phi : \mathbb{C}^{49} \rightarrow V_{128}$ is polynomial, and all its coefficients are integers as polynomials in θ .

Parameter block	Entries in order	Zero-based indices
Initial parameters	α, b, c, η	0, 1, 2, 3
Fourth product	$\mathbf{u}_4; \mathbf{v}_4$	4–6; 7–9
Fifth product	$\mathbf{u}_5; \mathbf{v}_5$	10–13; 14–17
Sixth product	$\mathbf{u}_6; \mathbf{v}_6$	18–22; 23–27
Seventh product	$\mathbf{u}_7; \mathbf{v}_7$	28–33; 34–39
Output	$e_0, \dots, e_8$	40–48

Table 1: The complete parameter ordering of (5).

Lemma 2. *For every $\theta \in \mathbb{C}^{49}$, one has $\Phi(\theta) \in X_7$. If $b \neq 2$, then $\Phi(\theta) \in \mathcal{P}_7$.*

Proof. The first two products are xx and $x^2(x^2 + \alpha x)$. For $b \neq 2$, set

$$g = b - 1, \quad \sigma = \frac{\eta - c + \alpha g}{b - 2}, \quad \rho = c - \sigma. \quad (6)$$

The identity

$$R = (Q + \rho x + gx^2)(Q + \sigma x + x^2) - gQ - \rho \sigma x^2 \quad (7)$$

therefore computes R with one additional product. Indeed, after the two displayed subtractions and the algebraic substitution $x^4 = Q - \alpha x^3$, the right-hand side is

$$Q^2 + (\rho + \sigma)xQ + (g + 1)x^2Q + (\rho + g\sigma - g\alpha)x^3.$$

The three relevant coefficients equal c, b, η , respectively. Every factor and subtraction in (7) uses only previously available polynomials. In particular, the expansion does not charge the unavailable monomial x^3 as a free operation.

Each of (3) costs one product, for a total of seven; (4) is free. Products and quotients of scalar parameters in (6) are allowed scalar coefficients. The degrees of $q_1, \dots, q_7$ are bounded by 2, 4, 8, 16, 32, 64, 128.

The set $\{b \neq 2\}$ is dense in $\mathbb{C}^{49}$, and the coefficient map Φ is polynomial even at $b = 2$. The inverse image $\Phi^{-1}(X_7)$ is closed and contains that dense open set; hence it is all of $\mathbb{C}^{49}$. No assertion that Φ is dominant onto X_7 is needed. $\square$

3 An exact contact and rank certificate

The frozen certificate file

`source/experiments/lower_full49_seed.json`

specifies a center $\theta_c \in \mathbb{Q}(i)^{49}$, an ordered list I of 48 distinct active indices, and a matrix $B \in \mathbb{Q}(i)^{48 \times 48}$. The center denominator is 10^{70} , and the matrix denominator is 10^{60} . The active list is reproduced in Appendix A; the sole fixed coordinate is $\theta_{48} = e_8$.

Let $z \in \mathbb{C}^{48}$ denote the active coordinates, in the order I . Keep the other coordinate fixed at its value in θ_c , and write $\iota(z) \in \mathbb{C}^{49}$ for the resulting full parameter vector. Define

$$F(z) = ([x^0]\Phi(\iota(z)), \dots, [x^{47}]\Phi(\iota(z)))^T - (0, \dots, 0, 1)^T, \quad J(z) = DF(z). \quad (8)$$

Let z_c be the active part of θ_c .

For a complex number use $|w|_1 = |\operatorname{Re} w| + |\operatorname{Im} w|$. For vectors use the maximum of these norms, and for matrices use the maximum row sum of entry norms. The latter bounds the induced vector norm; also $|uv|_1 \leq |u|_1|v|_1$.

Lemma 3 (Certified contact). *There is a point z_* with $\|z_* - z_c\| \leq 10^{-45}$ such that*

$$F(z_*) = 0, \quad \det J(z_*) \neq 0.$$

The associated parameter b is different from 2 throughout this ball. Thus the untruncated polynomial $\Phi(\iota(z_))$ has coefficients $0, \dots, 0, 1$ in degrees $0, \dots, 47$, and its selected coefficient differential has rank 48.*

Proof. The accompanying exact verifier evaluates the specified Gaussian-rational data and proves the following rational inequalities, with $r = 10^{-45}$:

$$\varepsilon := \|BF(z_c)\| < 10^{-69}, \quad \delta := \|I - BJ(z_c)\| < 10^{-57}, \quad (9)$$

$$\|B\| < 4000, \quad H < 2 \cdot 10^{12}, \quad (10)$$

$$q := \delta + \|B\| H r < 10^{-29}, \quad \varepsilon + qr < r. \quad (11)$$

It also checks

$$|\operatorname{Re}(b_c) - 2| > r. \quad (12)$$

Here H is a uniform aggregate second-derivative bound on the closed radius- r ball, as justified below. All these inequalities are evaluated using integers and rational numbers. Printed floating-point values are diagnostics, not certificate inputs.

For clarity, define the coefficient norm of a polynomial $a(x) = \sum_j a_j x^j$ by $\|a\|_{\text{coef}} = \sum_j |a_j|_1$. At every node of the full polynomial circuit propagate a triple (v, d, h) bounding its coefficient norm, the sum of the norms of its 48 first parameter derivatives, and the sum of the norms of all its 48^2 ordered second parameter derivatives. Addition adds these triples. Multiplication uses

$$(v, d, h) \cdot (w, e, k) = (vw, dw + ve, hw + 2de + vk). \quad (13)$$

An active scalar parameter has initial triple $(|\theta_{c,j}|_1 + r, 1, 0)$; a fixed scalar parameter has $(|\theta_{c,j}|_1, 0, 0)$; a fixed monomial has $(1, 0, 0)$. The product rule and submultiplicativity of $\|\cdot\|_{\text{coef}}$ prove these bounds inductively. The final value of h is the bound H in (10). In particular, it bounds the aggregate second derivatives of every output coefficient. Integration along a line segment in the convex ball gives

$$\|J(z) - J(z_c)\| \leq Hr. \quad (14)$$

Consider $G(z) = z - BF(z)$. Equations (11)–(14) give

$$\sup_{\|z - z_c\| \leq r} \|DG(z)\| \leq q < \frac{1}{2}, \quad \|G(z) - z_c\| \leq \varepsilon + qr < r.$$

The closed ball is complete, so Banach’s fixed point theorem supplies a fixed point z_* . Since $\delta < 1$, the matrix $BJ(z_c)$, and hence B , is invertible; thus $BF(z_*) = 0$ implies $F(z_*) = 0$. Moreover, $\|I - BJ(z_*)\| \leq q < 1$, so the Neumann series proves that $BJ(z_*)$, and therefore $J(z_*)$, is invertible. Finally, (12) keeps b different from 2 throughout the ball. $\square$

3.1 Exact arithmetic and independent reconstruction

The author-side check is implemented in `source/experiments/lower_full49_certify.py`, with Gaussian-integer helpers in `source/experiments/lower_contact_certify.py`. Value and Jacobian calculations may use $\mathbb{C}[x]/(x^{48})$, since this computes exactly the equations (8) and their derivatives. The argument itself uses the untruncated Φ in V_{128} ; no truncated polynomial is asserted to be a circuit output.

The parameter-degree bounds at $q_1, Q, R, q_4, q_5, q_6, q_7, \Phi$ are 0, 1, 2, 4, 8, 16, 32, 33. Consequently the value and derivative numerators can share denominator 10^{2310} . Derivative injection rescales numerators for differentiation with respect to the actual rational parameters, rather than their integer encodings. Matrix products, norms and asserted inequalities use Gaussian integers and exact rational arithmetic. The Hessian majorant traverses the full circuit and does not discard coefficients.

The independent implementation

`source/reviews/full49-degree47-fresh-check.py`

imports no author implementation. It evaluates all 129 coefficients and constructs all 49 derivative columns by a reverse pass through the product gates. It reproduces the exact preconditioned residual, inverse defect and inverse norm. Its separate positive-polynomial majorant, using the verified bound 4 for every parameter norm, gives

$$H_{\text{ind}} = 14225297189516125194807607250462624. \quad (15)$$

Although coarser, this bound still proves $q_{\text{ind}} < 6 \cdot 10^{-8}$ and $\varepsilon + q_{\text{ind}}r < r$ at the same radius. That checker also independently verifies $b \neq 2$. Thus the existence and rank claims do not depend on the author’s derivative or Hessian implementation. The numerical search and Decimal refinement are discovery records only; the frozen rational seed suffices for reproduction.

4 Full coefficient-space coverage

Proposition 4. *Every polynomial in V_{47} belongs to X_7 .*

Proof. Fix any monic polynomial $p(x) = \sum_{j=0}^{47} p_j x^j$ of degree 47. By Lemma 3, the selected coefficient map is locally invertible at z_* . The complex inverse function theorem therefore gives a holomorphic curve $z(t) \rightarrow z_*$, with the remaining parameter fixed, such that

$$[x^j]\Phi(\iota(z(t))) = p_j t^{47-j}, \quad j = 0, \dots, 47. \quad (16)$$

At $t = 0$ the prescribed coefficient vector is exactly $(0, \dots, 0, 1)$, so the local inverse applies for all sufficiently small t . In particular $z(t)$, and hence all coefficients of $\Phi(\iota(z(t)))$, remain bounded near zero.

For $t \neq 0$ set

$$f_t(x) = t^{-47}\Phi(\iota(z(t)))(tx). \quad (17)$$

Input substitution $x \mapsto tx$ and scalar output multiplication preserve the computational model and its closure. Thus $f_t \in X_7$. In fact, after shrinking the parameter neighbourhood if necessary, Lemma 2 and $b \neq 2$ give $f_t \in \mathcal{P}_7$. By (16), its coefficients through degree 47 are exactly those of p . For every $j = 48, \dots, 128$, its coefficient is t^{j-47} times a bounded holomorphic function, and therefore tends to zero. There are no coefficients above degree 128.

Consequently $f_t \rightarrow p$ in the full space V_{128} . An algebraic set over $\mathbb{C}$ is Euclidean closed, so $p \in X_7$. Scalar output multiplication gives every nonzero leading coefficient. Finally, any lower-degree polynomial is a limit of degree-47 polynomials, for example of $p + \epsilon x^{47}$, and closedness includes it as well. $\square$

This proof uses contact and scaling in addition to coefficient rank. Dominance of a coefficient projection alone would not establish the full-space closure assertion.

5 A self-contained matching upper bound

Lemma 5. *The variety X_7 is irreducible and has dimension at most 49.*

Proof. At each gate of a universal circuit, multiply two arbitrary linear combinations of all previously available polynomials, and take a final arbitrary linear combination. These coefficients form one affine parameter space, and the output map into V_{128} is polynomial. Circuits using fewer than seven products can be padded by zero products. Therefore its image closure is exactly X_7 , which is irreducible.

It suffices to bound the output dimension on the nonempty open subset where the successive product degrees are $2, 4, \dots, 128$. This subset is dense in the universal affine parameter space, and its image has the same closure. Let W_k be the space of available polynomials after k products; generically $\dim W_k = k + 2$.

After the first product, a free change of basis gives $W_1 = V_2$. After the second, subtract lower-degree terms and normalize the leading coefficient to obtain

$$W_2 = \text{span}\{1, x, x^2, Q\}, \quad Q = x^4 + \alpha x^3.$$

This family has dimension at most one. Denote by $W_2 W_2$ the linear span of its pairwise products. Then

$$W_2 W_2 = V_6 + \mathbb{C}Q^2, \quad \dim(W_2 W_2) = 8. \quad (18)$$

Indeed, products of V_2 span V_4 ; then xQ and x^2Q add degrees 5 and 6, while Q^2 adds degree 8. These span every pairwise product. Since $1 \in W_2$, the space W_2 is contained in $W_2 W_2$. The third product determines a line in the four-dimensional quotient $(W_2 W_2)/W_2$. Hence the family of possible W_3 spaces has dimension at most $1 + 3 = 4$.

For a fixed n -dimensional available space W containing 1, the next factors $(u, v) \in W^2$ determine $\text{span}(W, uv)$. On the generic set where $uv \notin W$ and both factors are nonconstant, each fiber of this space-valued map has dimension at least four. Specifically,

$$(u, v) \mapsto (au + b, cv + d), \quad a, c \in \mathbb{C} \setminus \{0\}, \quad b, d \in \mathbb{C}, \quad (19)$$

preserves the enlarged space, because the transformed product is $acuv + adu + bcv + bd$. Distinct quadruples in (19) give distinct pairs when u, v are nonconstant. The fiber-dimension bound thus shows that one product increases the dimension of a family of available spaces by at most $2n - 4$.

For the four remaining products, $n = 5, 6, 7, 8$, so the increments are 6, 8, 10, 12. The family of W_7 spaces therefore has dimension at most $4 + 6 + 8 + 10 + 12 = 40$. An arbitrary output in the nine-dimensional space W_7 adds at most nine dimensions. This gives $\dim X_7 \leq 40 + 9 = 49$.

The argument bounds a dense generic image and then its closure. It does not identify limiting multiplication spaces with products of limiting spaces, or rule out boundary circuits using degree profiles. $\square$

Proof of Theorem 1. Proposition 4 gives $d_7 \geq 47$. If $V_{48} \subseteq X_7$, then $\dim V_{48} = 49$ and Lemma 5 force $\dim X_7 = 49$. A proper closed subset of an irreducible variety has smaller dimension, so this containment would imply $X_7 = V_{48}$. But seven successive squarings compute x^{128} , which belongs to $\mathcal{P}_7$ and is outside V_{48} . Thus $V_{48} \not\subseteq X_7$, and every V_d with $d \geq 48$ is excluded. Hence $d_7 = 47$. $\square$

6 Reproduction and evidence scope

From the accompanying contribution directory

`references/webb-mf14-degree47-2026-09-17`

run

`python3 verification/verify.py`

The contribution README records the frozen files, hashes, exact outputs and separate review reports. The rational seed, rather than the numerical search trajectory, is the input needed to reproduce the existence and rank certificate. The two implementations check the finite arithmetic claims; the analytic and algebraic arguments in this manuscript have separate informal reviews. No Lean verification, proof-kernel checking, or external human peer review is claimed.

A Active-coordinate order

Relative to the zero-based ordering in Table 1, the certificate’s active coordinates are listed below in the order used for Jacobian columns and the rows of its preconditioner:

4	3	14	18	19	5	34	36	35	10	25	0
15	20	23	22	28	29	8	45	37	40	41	43
42	44	7	46	30	11	1	24	21	2	31	16
9	47	6	27	12	38	39	26	17	33	13	32.

All indices except 48 occur exactly once. The omitted coordinate e_8 is held at the exact Gaussian-rational value in the seed file. The preconditioner’s columns correspond to coefficient equations in increasing degree order $0, \dots, 47$.

References

- [1] E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3, Theorem 10 and Conjecture 13. <https://arxiv.org/html/2504.01500v3>.
- [2] M. J. Colbrook, *An exact degree-42 coverage certificate for seven polynomial multiplications*, 11 September 2026, Theorem 1. Archived in the OpenProblemsInNLA reference collection, [colbrook-matrix-functions-2026-09-11](#).
- [3] M. Webb, *Degree forty-four in the closure of seven-product polynomial evaluations: A negative resolution of MF-14*, 17 September 2026, Theorem 1 and Corollary 3. Archived in the OpenProblemsInNLA reference collection, [webb-mf14-degree44-2026-09-17](#).